Fashion-MNIST MLP Experiment Report

1. Task overview

This project uses a NumPy-implemented single-hidden-layer MLP to classify Fashion-MNIST.

2. Dataset and preprocessing

Fashion-MNIST contains 60,000 training images and 10,000 test images.

Each image is resized to 28x28 and flattened to a 784-dimensional vector.

The input is normalized by training-set pixel mean and standard deviation.

3. Model structure

Input dimension: 784

Hidden dimension: 512

Output dimension: 10

Activation: relu

4. Best hyperparameters

lr=0.2, lr_decay=0.95, weight_decay=0.001, epochs=50

Best validation accuracy: 0.9128

Test accuracy: 0.8940

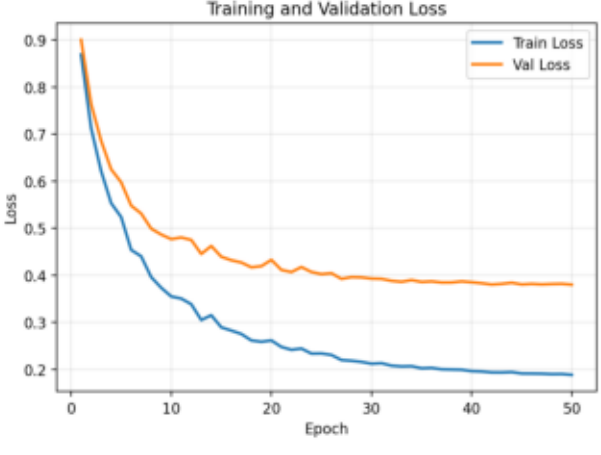
Test loss: 0.4414

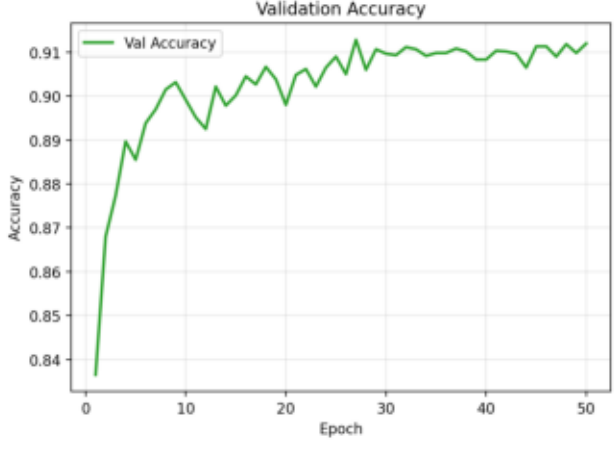
5. Submission links

GitHub repo: [Please insert your public repository link here]

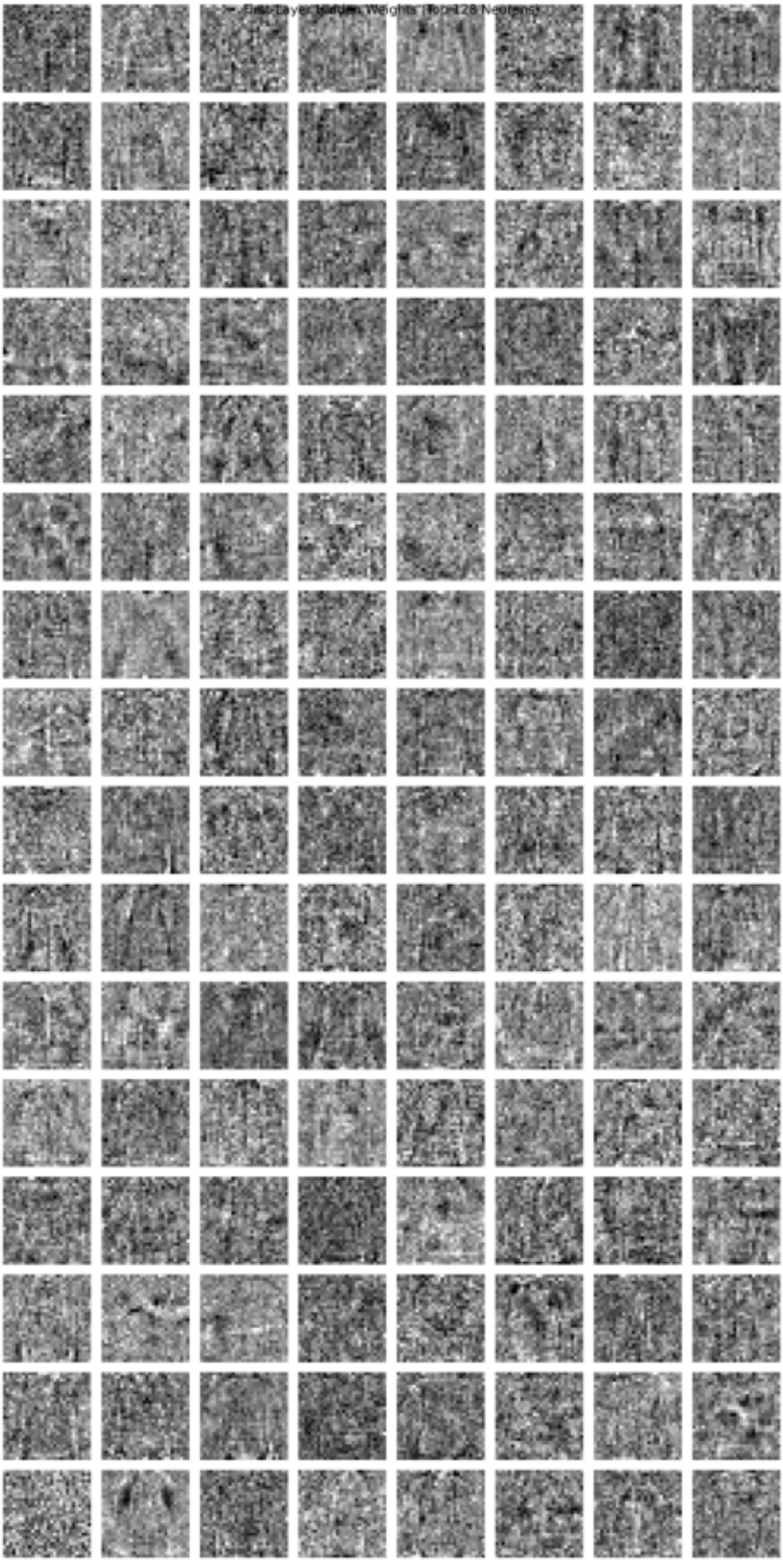
Model weights download: [Please insert your Google Drive or other download link here]

Training Curves

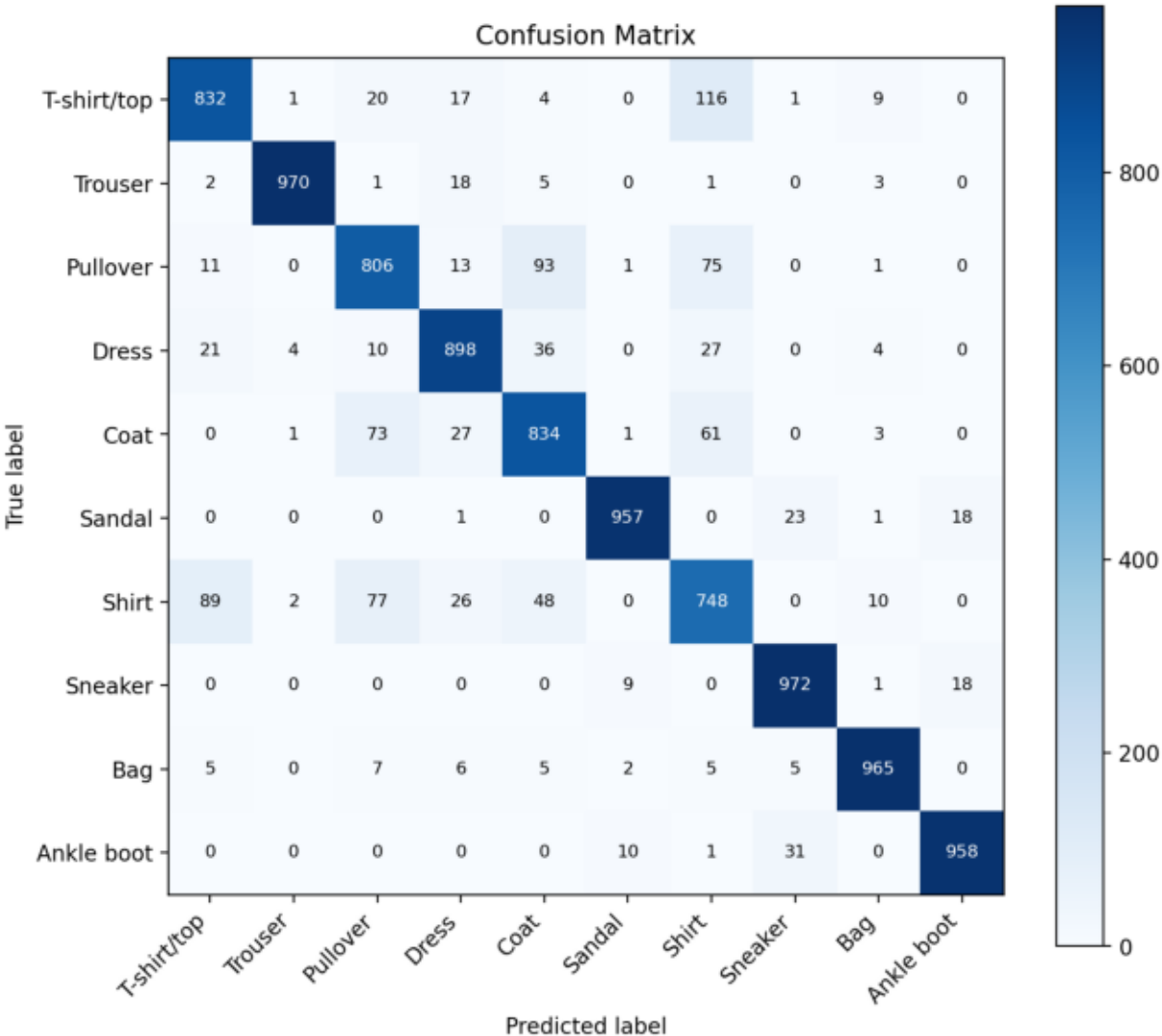




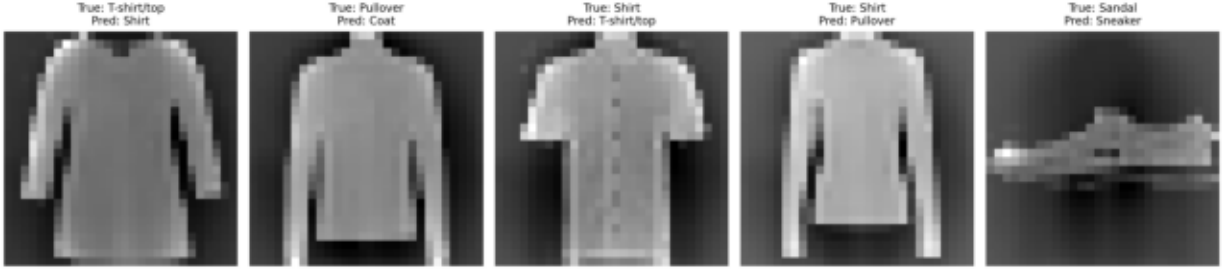
First-Layer Weight Visualization



Confusion Matrix



Misclassified Examples



6. Observations and Analysis

Training curves show that both training loss and validation loss decrease steadily, while validation accuracy rises and stabilizes near the best epoch. This indicates that the learning-rate schedule is effective and the model converges without severe instability.

The first-layer weights exhibit localized edge-like and contour-like patterns. Some hidden units respond to horizontal or vertical boundaries, while others emphasize diagonals, sleeve-like structures, shoe soles, and coarse clothing silhouettes. This suggests that the model learns low-level spatial templates before combining them for category discrimination.

Typical misclassifications occur among visually similar upper-body categories such as T-shirt/top, Shirt, Pullover, and Coat. These classes share overlapping outlines and differ mainly in subtle sleeve length, collar shape, or texture details, which are difficult to preserve after flattening a 28x28 grayscale image.

In the selected examples, Sandal and Sneaker can also be confused when the visible outline is narrow or when the sole structure dominates. This indicates that the model relies strongly on coarse silhouette information.

Overall, the best configuration achieves strong validation and test accuracy with a relatively simple MLP, but the remaining errors show that a fully connected model still has limited ability to capture fine local spatial structure compared with convolution-based models.

Selected error examples:
- T-shirt/top -> Shirt
- Pullover -> Coat
- Shirt -> T-shirt/top
- Shirt -> Pullover
- Sandal -> Sneaker